

AI Academy, National AI Center

Machine Learning

Team Contribution Report

Ensemble Methods: Boosting vs. Bagging

Team Name

Madagascar Penguins

Team Members

Murad, Bayram, Ferman, Mahmud

Semester:

Spring 2026

Version:

v1.0-final

Release Date:

June 29, 2026

Due Date:

July 17, 2026 at 23:59 (UTC+4)

Contribution Review

All implementations, experiments, revisions, and project updates were reviewed collaboratively by the entire team.

Issue #1 – Helper Modules and Utilities

Issue	Description and Files	Contributor
#2	Custom Metrics (evaluation.py)	Ferman
#3	Preprocessing (preprocessing.py)	Bayram
#37	Unsupervised Helper (unsupervised.py)	Murad
Main Issue Contributors		Bayram, Ferman, Murad, Mahmud
Review		Entire Team

Issue #4 – Implementation of Supervised Learning Models

Issue	Description and Files	Contributor
#5	Decision Tree (<code>decision_tree.py</code>)	Ferman
#6	Random Forest (<code>random_forest.py</code>)	Mahmud
#7	AdaBoost (<code>adaboost.py</code>)	Bayram
Main Issue Contributors		Bayram, Ferman, Mahmud
Review		Entire Team

Issue #8 – Implementation of Unsupervised Learning Models

Issue	Description and Files	Contributor
#9	K-Means (<code>kmeans.py</code>)	Murad
#10	PCA (<code>pca.py</code>)	Murad
#11	DBSCAN (<code>dbscan.py</code>)	Murad
Main Issue Contributor		Murad
Review		Entire Team

Issue #12 – Experiments

Issue	Description and Files	Contributor
#13	Baseline (<code>experiments_baseline.ipynb</code> , <code>experiments_baseline.py</code>)	Ferman
#14	AdaBoost Scaling (<code>adaboost_scaling_experiments_utils.py</code> , <code>adaboost_scaling.ipynb</code>)	Bayram
#15	Random Forest Scaling (<code>random_forest_scaling.ipynb</code> , <code>random_forest.py</code> , <code>decision_tree.py</code>)	Mahmud
#16	Head-to-Head Comparison (<code>head-to-head_5-Fold-CV.ipynb</code> , <code>experiments_head_to_head.py</code>)	Ferman
#17	Noise Robustness (<code>noise_robustness.ipynb</code> , <code>noise_helper.py</code>)	Bayram
#18	Bias-Variance Decomposition (<code>bias_variance_decomposition.ipynb</code> , <code>bias_variance_helper.py</code>)	Murad
#19	Unsupervised Analysis (<code>unsupervised_analysis.ipynb</code> , <code>unsupervised_helper.py</code>)	Murad
Main Issue Contributors		Bayram, Ferman, Murad, Mahmud
Review		Entire Team

Issue #20 – Revision

Issue	Description and Files	Contributor
#21	Final Report	Entire Team
#22	Contribution Statement (contribution_report.tex)	Murad
#23	Final Presentation	Entire Team
#51	README (README.md)	Ferman, Murad
#55	README Updates (README.md)	Ferman, Murad
Main Issue Contributors		Bayram, Ferman, Murad, Mahmud
Review		Entire Team

Issue #24 – Bonus

Issue	Description and Files	Contributor
#25	Continuous Integration Initial Setup (CI.yml)	Ferman
#27	t-SNE (EDA_of_four_datasets.ipynb)	Bayram
#28	Multiclass AdaBoost (adaboost.py)	Murad
Main Issue Contributors		Bayram, Ferman, Murad
Review		Entire Team

Final Contribution Statement

The project was completed collaboratively by the members of **Madagascar Penguins: Murad, Bayram, Ferman, and Mahmud**.

Each team member was responsible for specific implementation, experimentation, documentation, or revision tasks. All major changes, experimental results, source-code updates, and final deliverables were reviewed collectively by the entire team.